

TECHNICAL BULLETIN

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PRE-CATALYZED NADIC METHYL ANHYDRIDE (NMA 409)

Typical Properties

Properties	Values	Test Method
Molecular Weight (average)	188	
Specific gravity @ 20° C, g/cc	1.21 – 1.24	D-102 (ASTM D 4052)
Appearance	Clear dark green liquid	D-104 (Visual)
Viscosity at 25°C, cP	200 - 900	D-237

Applications

NADIC Methyl Anhydride 409 (NMA 409) curing agent formulated with an efficient latent proprietary catalyst that imparts high reactivity at process temperatures, while maintaining excellent pot life. NMA 409 is commonly used to cure epoxy resins in many challenging applications, including fiber reinforced composites used in high performance aerospace and military applications, as well as mechanically demanding applications. NMA 409 is ideal in processes such as filament winding, pultrusion and infusion of parts requiring excellent high temperature performance. The excellent electrical properties and high temperature performance make NMA-cured epoxies outstanding materials for encapsulating electronic components and circuits.

NMA 409 provides the following benefits:

- **Easy Handling:** Low viscosity liquid gives low viscosity mixes with various epoxy resins

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- **Convenience:** Eliminates the need to add small amounts of catalyst
- **Consistency:** Prevents potential batch-to-batch differences caused by errors in measuring and properly dispersing small amounts of catalyst
- **Low reactivity at room temperature:** Provides long working times and pot lives for mixes with epoxy resins
- **High reactivity at cure temperature:** Gives excellent cure response at cure temperatures, resulting in high crosslinking and relatively high Tg.
- **High Performance:** High heat distortion temperature (HDT) and glass transition temperature (Tg). Good thermal stability. Excellent electrical characteristics, including arc resistance. High hardness and impact strength

For more information on the use of anhydrides like NMA 409 as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical Company.

Epoxy Formulations

NMA 409 was formulated with a variety of epoxy resins. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 150°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

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Epoxy type	NMA 409, phr	Viscosity at 25 °C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	90	2150	16	168
Low Viscosity BPA Liquid	98	1490	19	174
Cycloaliphatic Epoxy	130	300	17	250
Epoxy Phenol Novolac	104	1640	18	164
Epoxy BPA Novolac	94	5930	18	183

1. Shyodu Hot Pot (100g at 100°C).

2. Cured 10°C/min to 300°C. Analyzed in DSC: 40°C/min to 300°C

Handling & Storage

It is recommended that NMA 409 is stored between the temperatures of 18° C (85° F) and 40° C (104° F) to avoid freezing, although the material is not affected by freezing. If material does freeze, thaw at a temperature below 40° C and mix well before using. Store in a dry place and keep container tightly closed to prevent absorption of atmospheric moisture or contamination. The presence of moisture could cause free acid to form in the anhydride.

NMA 409 will react with water to form diacids. This is normally undesirable, so NMA 409 should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for NMA 409, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants.

The data contained herein are furnished for information only and are believed to be reliable. This information is provided only as guidance and is not to be considered a warranty or quality specification. Dixie Chemical Company, Inc. makes no warranties (expressed or implied) as to its accuracy and assumes no liability in connection with any use of this product.

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Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

NMA 409 is available from Dixie Chemical Company, Inc.'s plant in Pasadena, Texas. Contact your Dixie Chemical Company representative for details.